Higher Complex Analysis. IV
Fractional linear transformations

Today?

In this lecture, we study a particular type of function: fractional linear transformations. These are easier to handle because we can use linear algebra to simplify computations.

Matrices and fractional linear transformations

Let M be a 2×2 complex matrix:

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We write T_M for the associated *fractional linear transformation*:

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First, we need to and will always assume that $(c, d) \neq (0, 0)$, otherwise the denominator is always 0.

Reducing the possibilities

If $\det M = 0$, then $(a, b) = \lambda(c, d)$, whence $az + b = \lambda(cz + d)$ and so $f(z) = \lambda$ for all $z \in \mathrm{Domain}(f)$. So fractional linear transformations associated to matrices with determinant 0 are essentially just constant functions. Henceforth, we assume that $\det M \neq 0$.

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Observe now that $T_{\lambda M} = T_M$ if $\lambda \neq 0$, and recall that $\det(\lambda M) = \lambda^2 \det(M)$. This means that when $\det M \neq 0$, if we set $M' = (\det M)^{-1/2} M$, then $\det M' = 1$ and $T_M = T_{M'}$. Therefore we may and shall restrict our attention to fractional linear transformations associated to matrices with determinant 1.

Domain and range

Henceforth we assume that $\det M = 1$. First we find the domain of T_M. If $c = 0$ and $d \neq 0$, then the denominator is never 0, so $T_M(z)$ is defined for all a, that is, $\mathrm{Domain}(T_M) = \mathbb{C}$; otherwise, the denominator is nonzero when $z \neq -d/c$, whence $\mathrm{Domain}(T_M) = \mathbb{C} \setminus \{-d/c\}$.

Inverse transformations

Now suppose that $w \in \text{Range}(T_M)$. Then

$$\begin{aligned} w = \frac{az+b}{cz+d} \quad &\Longrightarrow \quad wcz + wd = az + b \\ &\Longrightarrow \quad z(wc - a) = b - wd, \end{aligned}$$

and as long as $wc \neq a$, this in turn implies that

$$z = \frac{b - wd}{wc - a} = \frac{(-d)w + b}{cw + (-a)} = \frac{dw - b}{(-c)w + a} = T_{M'}(w)\,,$$

where

$$M' = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Then M' is exactly M^{-1}, since $\det M = 1$.

Range and domain

We conclude that

$$\mathrm{Range}(T_M) = \begin{cases} \mathbb{C} \setminus \{a/c\} & \text{if } c \neq 0 \\ \mathbb{C} & \text{if } c = 0. \end{cases}$$

Simplifying the range and domain

Our discussion of the domain and range had several cases; we can simplify the statements by enlarging the set of complex numbers by adding ∞. Indeed, we define

$$f\left(-\frac{d}{c}\right) = \lim_{z \to -d/c} \frac{az+b}{cz+d} = \infty \quad \text{and} \quad f(\infty) = \lim_{z \to \infty} \frac{az+b}{cz+d} = \frac{a}{c}$$

(we will define limits formally in the next lecture), and now we can just write

$$T_M : \mathbb{C} \cup \{\infty\} \to \mathbb{C} \cup \{\infty\}.$$

The Riemann sphere

We often call $\mathbb{C} \cup \{\infty\}$ the Riemann sphere, and write it S: we imagine a unit sphere in three dimensions with centre at 0. We can define a function $\sigma : \mathbb{C} \to S$ geometrically by joining a point p in the plane to the north pole n of the sphere by a straight line. The line will cut the sphere at n and at one other point, which we call $\sigma(p)$. Then we may think of n as being $\sigma(\infty)$. The function σ is called *stereographic projection*.

Exercise 1

Suppose that $M, N \in M_{2,2}(\mathbb{C})$. Show that $T_M T_N = T_{MN}$ (on the Riemann sphere). Deduce that the transformation T_M is bijective and that $(T_M)^{-1} = T_{M^{-1}}$.

Answer.

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Answer. Write $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $N = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$. Then we compute

$$MN = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

Answer to Exercise 1

Further,

$$T_M T_N(z) = \frac{aT_N(z) + b}{cT_N(z) + d} = \frac{a\dfrac{ez+f}{gz+h} + b}{c\dfrac{ez+f}{gz+h} + d}$$

$$= \frac{a(ez+f) + b(gz+h)}{c(ez+f) + d(gz+h)} = \frac{(ae+bg)z + (af+bh)}{(ce+dg)z + (cf+dh)}.$$

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$$= \frac{a(ez+f)+b(gz+h)}{c(ez+f)+d(gz+h)} = \frac{(ae+bg)z+(af+bh)}{(ce+dg)z+(cf+dh)}.$$

This formula holds for all $z \in \mathbb{C}$ for which none of the denominators is 0, and then extends to all $z \in S$ by continuity. It follows that

$$T_M T_N(z) = T_{MN}(z) \qquad \forall z \in S.$$

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$$T_M T_N(z) = T_{MN}(z) \qquad \forall z \in S.$$

In particular, $T_M T_{M^{-1}} = T_I$, and T_I is the identity mapping. Hence T_M is invertible, with inverse $T_{M^{-1}}$. $\triangle$

Factorisations of fractional linear transformations

Matrices may be factorised, and hence fractional linear transformations may be factorised too.

Theorem

Every 2×2 complex matrix with determinant 1 may be written as a product of at most three matrices of the following special types:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \qquad \textit{and} \qquad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Proof. Consider the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We consider three cases, according to whether any of a or c is 0. Since the determinant is 1, both cannot be 0.

If $c = 0$, the matrix itself is of the desired form.

Proof of theorem

Now suppose that $a = 0$. Then

$$\begin{aligned}
\begin{pmatrix} 0 & b \\ c & d \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 0 & b \\ c & d \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & b \\ c & d \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -c & -d \\ 0 & b \end{pmatrix},
\end{aligned}$$

which is of the desired form.

Proof of theorem (continued)

Finally, if neither a nor c is 0, then we take $x = a/c$, and write

$$\begin{aligned}\begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a - cx & b - dx \\ c & d \end{pmatrix};\end{aligned}$$

since $a - cx = 0$, combining with the previous case shows that the factorisation holds in this case too. $\square$

Proof of theorem (continued)

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This factorisation simplifies a number of arguments.

Fractional linear transformations, lines and circles

Theorem

Let T_M be a fractional linear transformation. Then the image of a line under T_M is a line or a circle, and the image of a circle under T_M is also a line or a circle.

Proof. If M is of the form $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$, then T_M is an affine transformation, and the theorem holds in this case. Otherwise, T_M is composed of some affine transformations and the *inversion* map $z \mapsto -1/z$; it therefore suffices to treat the inversion map.

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We may write the equation of the circle with centre c and radius r in the form $|z - c|^2 = r^2$. This is the same as

$$|z|^2 - 2\,\mathrm{Re}(z\bar{c}) + |c|^2 - |r|^2 = 0.$$

Proof of theorem

$$|z|^2 - 2\,\mathrm{Re}(z\bar{c}) + |c|^2 - |r|^2 = 0.$$

If we set $w = -1/z$, then we find that

$$\left|\frac{-1}{w} - c\right|^2 = r^2,$$

whence

$$|wc + 1|^2 = r^2|w|^2,$$

and

$$\left(|c|^2 - r^2\right)|w|^2 + 2\,\mathrm{Re}(wc) + 1 = 0,$$

which is the equation of a circle, unless $|c|^2 = r^2$, when we get the equation of a straight line.

Proof of theorem. 2

The argument to show that the inversion mapping sends straight lines to straight lines or circles is similar, and we omit it; the starting point is that the equation of a straight line may also be written

$$|z - p| = |z - q|. \qquad \square$$

Exercise 2

Show that the fractional linear transformation $T : z \mapsto \dfrac{z-i}{z+i}$ sends the upper half plane onto the unit disc $\{w \in \mathbb{C} : |w| < 1\}$.
Find the image under T of the wedge
$\{z \in \mathbb{C} : \varphi < \mathrm{Arg}(z) < \pi - \varphi\}$.

Answer.

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Answer. We know that T is invertible and may check easily that T^{-1} is the fractional linear transformation
$w \mapsto -i\dfrac{w+1}{w-1}$.

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$w \mapsto -i\dfrac{w+1}{w-1}$.

Suppose that $z = x + iy$, where $y > 0$. Then

$$|Tz|^2 = \left|\frac{x+iy-i}{x+iy+i}\right|^2 = \frac{x^2+(y-1)^2}{x^2+(y+1)^2} < 1,$$

so T maps into the unit disc.

Answer to Exercise 2

On the other hand, if $|w| < 1$, then

$$T^{-1}w = -i\frac{w+1}{w-1} = -i\frac{(w+1)(\bar{w}-1)}{(w-1)(\bar{w}-1)}$$
$$= \frac{i(w-\bar{w}) + i(1-w\bar{w})}{|w-1|^2},$$

so $\mathrm{Im}(T^{-1}w) = (1-|w|^2)/(|1-w|^2) > 0$; this implies that w is in the image of the upper half plane under T, so T maps onto the unit disc. Further, $T(0) = -1$ and $T(\infty) = 1$.

Answer to Exercise 2 (continued)

Let R_θ denote the ray $\{re^{i\theta} : r \in \mathbb{R}^+\}$. Each ray R_θ is mapped into the part of a line or a circle that lies inside the unit disc and starts at -1 and end at 1.

Answer to Exercise 2 (continued)

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$$\begin{aligned} Te^{i\theta} &= \frac{e^{i\theta} - i}{e^{i\theta} + i} = \frac{e^{i\theta} - e^{i\pi/2}}{e^{i\theta} + e^{i\pi/2}} \\ &= \frac{e^{i\theta/2 - i\pi/4} - e^{i\pi/4 - i\theta/2}}{e^{i\theta/2 - i\pi/4} + e^{i\pi/4 - i\theta/2}} = i\tan(\theta/2 - \pi/4). \end{aligned}$$

It follows that the image of the ray is A_θ, the arc of the circle with centre $(0, \tan(\theta))$ between $(-1, 0)$ and $(1, 0)$.

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Hence the image of the wedge is the "eye-shaped" region between the arcs A_φ and $A_{\pi-\varphi}$. $\triangle$

Exercise 3

Suppose that $M \in M_{2,2}(\mathbb{R})$ (all the coefficients of the matrices are real) and $\det(M) = 1$. Show that if $\mathrm{Im}(z) > 0$, then $\mathrm{Im}(T_M z) > 0$. Deduce that T_M maps the upper half plane $\{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$ onto itself bijectively.

Answer.

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Answer. Suppose that $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \mathbb{R}$.

Then

$$T_M(z) = \frac{az+b}{cz+d} = \frac{(az+b)(cz+d)^-}{(cz+d)(cz+d)^-}$$
$$= \frac{(az+b)(c\bar{z}+d)}{|cz+d|^2} = \frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz+d|^2}.$$

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Suppose that $z = x + iy$, where $y > 0$. Then ...

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$$\mathrm{Im}(T_M(z)) = \mathrm{Im}\left(\frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz + d|^2}\right)$$

Answer to Exercise 3

$$\begin{aligned}\mathrm{Im}(T_M(z)) &= \mathrm{Im}\Big(\frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz+d|^2}\Big) \\ &= \frac{\mathrm{Im}\big(ac|z|^2 + adz + bc\bar{z} + bd\big)}{|cz+d|^2}\end{aligned}$$

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$$\begin{aligned}
\mathrm{Im}(T_M(z)) &= \mathrm{Im}\Big(\frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz+d|^2}\Big) \\
&= \frac{\mathrm{Im}\big(ac|z|^2 + adz + bc\bar{z} + bd\big)}{|cz+d|^2} \\
&= \frac{\mathrm{Im}(ad(x+iy) + bc(x-iy))}{|cz+d|^2}
\end{aligned}$$

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&= \frac{\operatorname{Im}((ad+bc)x + i(ad-bc)y)}{|cz+d|^2} \\
&= \frac{(ad-bc)y}{|cz+d|^2} \\
&= \frac{y}{|cz+d|^2} > 0.
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&= \frac{(ad-bc)y}{|cz+d|^2} \\
&= \frac{y}{|cz+d|^2} > 0.
\end{aligned}
$$

Thus $\operatorname{Im}(T_M(z)) > 0$, as required.

Answer to Exercise 3 (continued)

From a previous exercise, T_M is bijective. It remains to show that if $\mathrm{Im}(w) > 0$, then the complex number z such that $T_M(z) = w$ satisfies $\mathrm{Im}(z) > 0$. But $T_M(z) = w$ is the same as $z = (T_M)^{-1}w = T_{M^{-1}}w$; by linear algebra, $M^{-1} \in M_{2,2}(\mathbb{R})$ and $\det(M^{-1}) = 1$. The first part of this answer, applied to M^{-1} rather than M, shows that $\mathrm{Im}(T_{M^{-1}}w) > 0$, and thus $\mathrm{Im}(z) > 0$, as required. $\triangle$

Further exercises

Exercise

Suppose that M is a 2×2 complex matrix of determinant 1, and that T_M maps the upper half plane onto itself. Show that M is a real matrix.

Exercise

Find a characterisation of all 2×2 complex matrices M of determinant 1 such that T_M maps the unit disc onto itself.

Exercise

Suppose that T_M is a fractional linear transformation that maps the circle C_1 onto the circle C_2. When does T_M maps the interior of C_1 onto the interior of C_2?